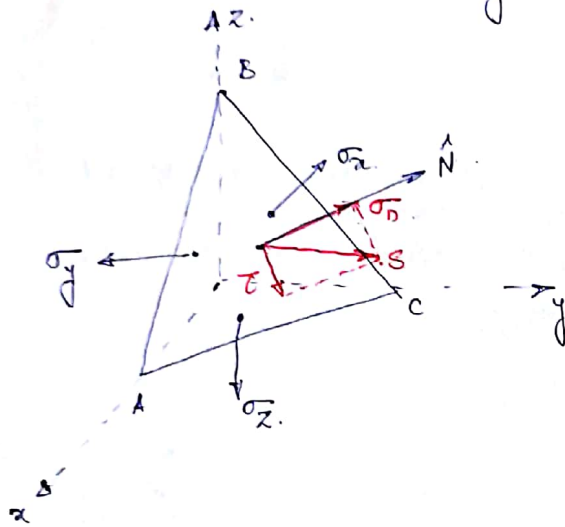


04/04/19

①

Maximum shearing stress

Assume that the x, y and z axes are the principal axes.



Let l, m, n be the direction cosines of the plane ABC.

X, Y, Z are the components of the resultant stress along the coordinate axes.

The total stress

$$X = \sigma_x l, Y = \sigma_y m, Z = \sigma_z n.$$

The square of the resultant stress is.

$$S^2 = X^2 + Y^2 + Z^2 = \sigma_x^2 l^2 + \sigma_y^2 m^2 + \sigma_z^2 n^2. \quad \text{--- (1)}$$

The normal stress on the plane ABC is.

$$\sigma_n = \sigma_x l^2 + \sigma_y m^2 + \sigma_z n^2 + 2\tau_{xy} lm + 2\tau_{yz} mn + 2\tau_{xz} ln.$$

$$\Rightarrow \sigma_n = \sigma_x l^2 + \sigma_y m^2 + \sigma_z n^2. \quad \text{--- (2)}$$

But we know that $S^2 = \sigma_n^2 + \tau^2$, where τ is the shearing stress on plane ABC. (*)

$$\therefore \tau^2 = S^2 - \sigma_n^2 = \sigma_x^2 l^2 + \sigma_y^2 m^2 + \sigma_z^2 n^2 - (\sigma_x l^2 + \sigma_y m^2 + \sigma_z n^2)^2 \quad \text{--- (3)}$$

$$\text{We also know that } l^2 + m^2 + n^2 = 1. \quad \text{--- (4)}$$

(2)

$$\therefore \tau^2 = \sigma_x^2 l^2 + \sigma_y^2 m^2 + \sigma_z^2 (1-l^2-m^2) - \left[\sigma_x l^2 + \sigma_y m^2 + \sigma_z (1-l^2-m^2) \right]^2$$

$$\Rightarrow \tau^2 = (\sigma_x^2 - \sigma_z^2) l^2 + (\sigma_y^2 - \sigma_z^2) m^2 + \sigma_z^2 - \left[(\sigma_x - \sigma_z) l^2 + (\sigma_y - \sigma_z) m^2 + \sigma_z \right]^2 \quad \text{--- (5)}$$

The direction cosines of the plane for which the shear stress is maximum can be obtained from

$$\frac{\partial \tau^2}{\partial l} = 0 \quad \text{and} \quad \frac{\partial \tau^2}{\partial m} = 0$$

$$\frac{\partial \tau^2}{\partial l} = 0 \Rightarrow 2l (\sigma_x^2 - \sigma_z^2) - 2 \left[(\sigma_x - \sigma_z) l^2 + (\sigma_y - \sigma_z) m^2 + \sigma_z \right] \times 2l (\sigma_x - \sigma_z) = 0$$

$$\Rightarrow l \left[(\sigma_x + \sigma_z) - 2 \left\{ (\sigma_x - \sigma_z) l^2 + (\sigma_y - \sigma_z) m^2 + \sigma_z \right\} \right] = 0$$

$$\Rightarrow l \left[(\sigma_x - \sigma_z) l^2 + (\sigma_y - \sigma_z) m^2 - \frac{(\sigma_x - \sigma_z)}{2} \right] = 0 \quad \text{--- (6a)}$$

Similarly, $\frac{\partial \tau^2}{\partial m} = 0$

$$\Rightarrow m \left[(\sigma_x - \sigma_z) l^2 + (\sigma_y - \sigma_z) m^2 - \frac{(\sigma_y - \sigma_z)}{2} \right] = 0 \quad \text{--- (6b)}$$

The trivial solⁿ to (6a) and (6b) is.

$$l = m = 0 \Rightarrow n = \pm 1 \quad (\because l^2 + m^2 + n^2 = 1) \quad \text{--- (*)}$$

The non-trivial solⁿ can be obtained by sub. $l=0$ in (6b)

$$\Rightarrow (\sigma_y - \sigma_z) m^2 - \frac{\sigma_y - \sigma_z}{2} = 0$$

$$\Rightarrow m = \pm \frac{1}{\sqrt{2}}$$

$$\Rightarrow l=0, m = \pm \frac{1}{\sqrt{2}} \quad \text{and} \quad n = \pm \frac{1}{\sqrt{2}} \quad \text{--- (**)}$$

Also by sub. $m=0$ in (6a).

$$\Rightarrow (\sigma_1 - \sigma_2) l^2 - \frac{\sigma_1 - \sigma_2}{2} = 0$$

$$\Rightarrow l = \pm \frac{1}{\sqrt{2}}$$

$$\Rightarrow \underline{l = \pm \frac{1}{\sqrt{2}}, m=0, n = \pm \frac{1}{\sqrt{2}}} \quad \text{--- (xx) ---}$$

Repeating the above calculation by eliminating l and m from (3).
we can find that τ is stationary when.

$$l=0, m=0, n=\pm 1.$$

$$l=0, m=\pm 1, n=0.$$

$$l=\pm 1, m=0, n=0$$

planes are $\perp$ to. Principal axes.
shearing stress is zero
(minimum).

$$l=0, m=\pm \frac{1}{\sqrt{2}}, n=\pm \frac{1}{\sqrt{2}}$$

$$l=\pm \frac{1}{\sqrt{2}}, m=0, n=\pm \frac{1}{\sqrt{2}}$$

$$l=\pm \frac{1}{\sqrt{2}}, m=\pm \frac{1}{\sqrt{2}}, n=0$$

planes are along one of
the principal axes and
bisect the other two.
(maximum).

Maximum shear stress

[Using eq (3)]

$$\text{For } l=0, m=\pm \frac{1}{\sqrt{2}}, n=\pm \frac{1}{\sqrt{2}} \quad , \quad \tau_{\max} = \frac{\sigma_y - \sigma_z}{2}$$

$$\text{For } l=\pm \frac{1}{\sqrt{2}}, m=0, n=\pm \frac{1}{\sqrt{2}} \quad , \quad \tau = \frac{\sigma_1 - \sigma_2}{2}$$

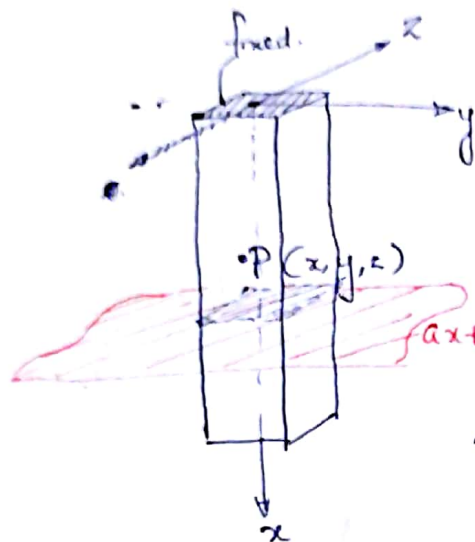
$$\text{For } l=\pm \frac{1}{\sqrt{2}}, m=\pm \frac{1}{\sqrt{2}}, n=0 \quad , \quad \tau = \frac{\sigma_1 - \sigma_3}{2}$$

Maximum shear stress is given by the largest of the above
three, equal to ~~the~~ half the difference between the
largest and the smallest principal stresses.

(4)

Homogeneous deformation

Consider a prismatic bar under tension. One end of the bar is fixed.



Normal strain $\epsilon_x = \epsilon$

Poisson effect. $\begin{cases} \epsilon_y = -\nu \epsilon \\ \epsilon_z = -\nu \epsilon \end{cases}$

$ax + by + cz = d$

$\therefore$ Components of displacement of a point $P(x, y, z)$ are

$u = \epsilon x$, $v = -\nu \epsilon y$, $w = -\nu \epsilon z$

The coordinates of point P after deformation

$x' = x + u = x(1 + \epsilon)$

$y' = y + v = y(1 - \nu \epsilon)$

and $z' = z + w = z(1 - \nu \epsilon)$

#1 Consider a plane in the undeformed body defined by

$ax + by + cz = d$

After deformation : $a(x' - u) + b(y' - v) + c(z' - w) = d$

$\Rightarrow a \frac{x'}{1 + \epsilon} + b \frac{y'}{1 - \nu \epsilon} + c \frac{z'}{1 - \nu \epsilon} = d$

The same points remain in one plane.

#2 Consider two ^{parallel} planes in the undeformed body

$$P_1 : ax + by + cz = d_1$$

$$P_2 : ax + by + cz = d_2$$

After deformation :

$$P_1^* : \frac{ax'}{1+\epsilon} + \frac{by'}{1-\nu\epsilon} + \frac{cz'}{1-\nu\epsilon} = d_1$$

$$P_2^* : \frac{ax'}{1+\epsilon} + \frac{by'}{1-\nu\epsilon} + \frac{cz'}{1-\nu\epsilon} = d_2$$

The planes P_1^* and P_2^* (new planes: P_1^* and P_2^*) remain parallel.

#3 Consider a spherical surface in the bar before deformation

$$x^2 + y^2 + z^2 = r^2$$

After deformation :

$$\frac{x'^2}{r^2(1+\epsilon)^2} + \frac{y'^2}{r^2(1-\nu\epsilon)^2} + \frac{z'^2}{r^2(1-\nu\epsilon)^2} = 1$$

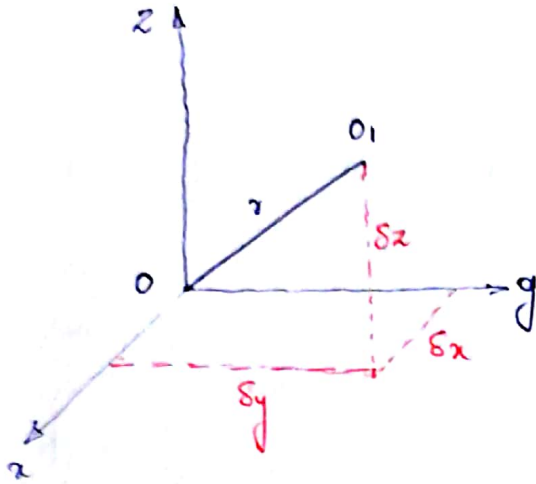
The sphere deforms into an ellipsoid, with semi-axes.

$$r(1+\epsilon), r(1-\nu\epsilon) \text{ and } r(1-\nu\epsilon).$$

This kind of deformation in which ① planes and straight lines remain plane and straight after deformation
 ② parallel planes and parallel straight lines remain parallel after deformation and ③ a sphere becomes an ellipsoid.
 is called homogeneous deformation.

Strain at a point

Consider a small linear element of length r ,
with direction cosines l, m and n .



Let the origin is at O.
The coordinates of point O_1
are.

$$\delta x = rl, \delta y = rm \\ \text{and } \delta z = rn.$$

Let u, v, w represent the components of displacement
of point O (the coordinate system is attached to point O)

The corresponding displ. of point O_1 are

$$u_1 = u + \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y + \frac{\partial u}{\partial z} \delta z$$

$$v_1 = v + \frac{\partial v}{\partial x} \delta x + \frac{\partial v}{\partial y} \delta y + \frac{\partial v}{\partial z} \delta z.$$

$$w_1 = w + \frac{\partial w}{\partial x} \delta x + \frac{\partial w}{\partial y} \delta y + \frac{\partial w}{\partial z} \delta z.$$

In the above equation only the first order terms of
 $\delta x, \delta y$ and δz are considered. Higher order terms are
small and can be neglected.

The coordinates of point O_1 after deformation: δ

$$\delta x + u_1 - u = \delta x + \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y + \frac{\partial u}{\partial z} \delta z$$

$$\delta y + v_1 - v = \delta y + \frac{\partial v}{\partial x} \delta x + \frac{\partial v}{\partial y} \delta y + \frac{\partial v}{\partial z} \delta z.$$

$$\delta z + w_1 - w = \delta z + \frac{\partial w}{\partial x} \delta x + \frac{\partial w}{\partial y} \delta y + \frac{\partial w}{\partial z} \delta z.$$

Since the coordinates are linear functions of the initial coordinates $\delta x, \delta y, \delta z$, the deformation is homogeneous.

The Elongation of the element OO_1

Square of the length of OO_1 after

$$\text{deformation} = (\delta x + u_1 - u)^2 + (\delta y + v_1 - v)^2 + (\delta z + w_1 - w)^2$$

If ϵ is the unit elongation of the element.

Square of the length of OO_1 after

$$\text{deformation} = r^2 (1 + \epsilon)^2.$$

$$\Rightarrow r^2 (1 + \epsilon)^2 = \left(\delta x + \frac{\partial u}{\partial x} \delta x + \frac{\partial u}{\partial y} \delta y + \frac{\partial u}{\partial z} \delta z \right)^2 + \left(\delta y + \frac{\partial v}{\partial x} \delta x + \frac{\partial v}{\partial y} \delta y + \frac{\partial v}{\partial z} \delta z \right)^2 + \left(\delta z + \frac{\partial w}{\partial x} \delta x + \frac{\partial w}{\partial y} \delta y + \frac{\partial w}{\partial z} \delta z \right)^2$$

Substituting for $\delta x, \delta y, \delta z$ and dividing by r^2 ,

$$(1 + \epsilon)^2 = \left[l \left(1 + \frac{\partial u}{\partial x} \right) + m \frac{\partial u}{\partial y} + n \frac{\partial u}{\partial z} \right]^2 + \left[l \frac{\partial v}{\partial x} + m \left(1 + \frac{\partial v}{\partial y} \right) + n \frac{\partial v}{\partial z} \right]^2 + \left[l \frac{\partial w}{\partial x} + m \frac{\partial w}{\partial y} + n \left(1 + \frac{\partial w}{\partial z} \right) \right]^2$$

In the above eqn.

$$\left[l(1 + \frac{\partial u}{\partial x}) + \frac{\partial u}{\partial x} m + \frac{\partial u}{\partial y} n \right]^2 = l^2 (1 + \frac{\partial u}{\partial x})^2 + 2lm \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + 2mn \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial y}$$

($\therefore$ higher order terms of small quantities)

$$\approx l^2 \left[1 + 2 \frac{\partial u}{\partial x} + \left(\frac{\partial u}{\partial x} \right)^2 + 2lm \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial y} \right]$$

$$\approx l^2 \left[1 + 2 \frac{\partial u}{\partial x} + 2lm \frac{\partial u}{\partial x} + 2ln \frac{\partial u}{\partial y} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} \right]$$

Similarly,

$$\left[l \frac{\partial v}{\partial x} + m(1 + \frac{\partial v}{\partial y}) + n \frac{\partial v}{\partial z} \right]^2 \approx m^2 \left[1 + 2 \frac{\partial v}{\partial y} + 2lm \frac{\partial v}{\partial x} + 2mn \frac{\partial v}{\partial z} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial z} \right]$$

$$\approx m^2 \left[1 + 2 \frac{\partial v}{\partial y} + 2ln \frac{\partial v}{\partial z} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial z} \right]$$

$$\Rightarrow (1 + \epsilon)^2 = \underbrace{l^2 + m^2 + n^2}_{1} + 2l \frac{\partial u}{\partial x} + 2m \frac{\partial v}{\partial y} + 2n \frac{\partial w}{\partial z} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \frac{\partial w}{\partial z} + \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial v}{\partial y} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial w}{\partial z} + \frac{\partial w}{\partial z} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial w}{\partial z} + \frac{\partial w}{\partial z} \frac{\partial v}{\partial y}$$

$$1 + 2\epsilon + \epsilon^2 = 1 + 2\epsilon + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \frac{\partial w}{\partial z} + \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial v}{\partial y} \frac{\partial u}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial w}{\partial z} + \frac{\partial w}{\partial z} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \frac{\partial w}{\partial z} + \frac{\partial w}{\partial z} \frac{\partial v}{\partial y}$$

ϵ^2 (small term)

$$\Rightarrow \epsilon = l^2 \frac{\partial u}{\partial x} + m^2 \frac{\partial v}{\partial y} + n^2 \frac{\partial w}{\partial z} + 2lm \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) + 2mn \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right) + 2ln \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right).$$

We know that

$$\frac{\partial u}{\partial x} = \epsilon_x, \quad \frac{\partial v}{\partial y} = \epsilon_y, \quad \frac{\partial w}{\partial z} = \epsilon_z.$$

$$\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = \gamma_{xy}, \quad \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = \gamma_{yz}, \quad \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} = \gamma_{xz}.$$

or write

$\Rightarrow$ Elongation of a linear element is.

$$\epsilon = l^2 \epsilon_x + m^2 \epsilon_y + n^2 \epsilon_z + 2lm \gamma_{xy} + 2mn \gamma_{yz} + 2ln \gamma_{xz}.$$

$$\text{or } \epsilon = l^2 \epsilon_x + m^2 \epsilon_y + n^2 \epsilon_z + 2lm \epsilon_{xy} + 2mn \epsilon_{yz} + 2ln \epsilon_{xz}$$

For homogeneous deformation, u, v, w are linear functions of coordinates x, y and z . $\Rightarrow$ The strains (refer to the eqn of components of strain above) are constant over the volume of the body.

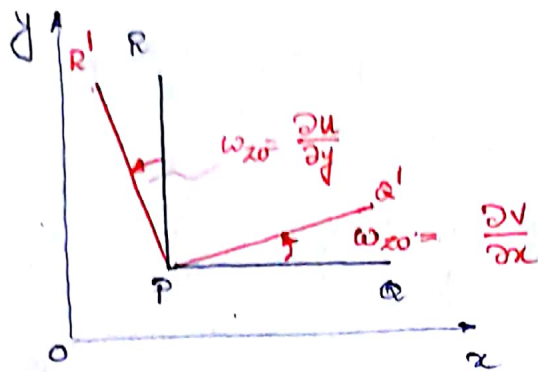
Strain tensor:

$$[\epsilon] = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & \epsilon_{xz} \\ \epsilon_{yx} & \epsilon_{yy} & \epsilon_{yz} \\ \epsilon_{zx} & \epsilon_{zy} & \epsilon_{zz} \end{bmatrix}$$

$$\epsilon_{xy} = \frac{1}{2} \gamma_{xy}, \quad \epsilon_{yz} = \frac{1}{2} \gamma_{yz}, \quad \epsilon_{xz} = \frac{1}{2} \gamma_{xz}.$$

$$\epsilon_{xy} = \epsilon_{yx}, \quad \epsilon_{yz} = \epsilon_{zy}, \quad \epsilon_{xz} = \epsilon_{zx} \quad (\text{Cross shears are equal})$$

Rotation



If the element PQR undergoes a pure rigid body rotation through a small angle about P,

$$\omega_z = \frac{\partial v}{\partial x} = - \frac{\partial u}{\partial y} \quad (\text{taking ccw as +ve})$$

The negative sign is that due to a change in u along -ve x dir as y increases. A positive $\frac{\partial u}{\partial y}$ will look like

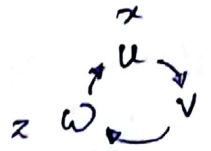
look like $\frac{\partial u}{\partial y} (+ve)$.

Note that no strain occurs during this rigid body rotation. One can define the rotation of the sq element by the average of the sum of rotations of x and y elements (PQ and QR).

$$\omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

Similarly, for rotations about the x and y °

$$\omega_y = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial u}{\partial z} \right)$$



$$\omega_x = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial y} \right)$$

and

$$\omega_z = \frac{1}{2} \left(\frac{\partial w}{\partial x} - \frac{\partial v}{\partial y} \right)$$